

Dental Clinic Software

India-Wide Product Requirements & Feature Blueprint

Product vision: Build a dental-specific clinic operating system connecting patient management, diagnosis, treatment, consent, appointments, inventory, laboratory, billing, follow-up, consultant management and clinic profitability.

1. Patient Master Profile

- Automatically generated Patient ID.
- Name, DOB/age, gender, phone/WhatsApp, email, address, occupation, emergency contact.
- Referral source and preferred language/communication preference.
- Medical history: allergies, diabetes, hypertension, cardiac conditions, pregnancy, medications, bleeding disorders, surgeries and ASA classification.
- Prominent medical alerts before treatment.

2. Dental Case Sheet & Odontogram

- Interactive permanent and primary-tooth chart.
- Record caries, restorations, missing teeth, crowns, RCT, extraction, implants, mobility, periodontal findings and trauma.
- Every tooth should have a lifetime clinical history.
- Case-level documentation linked to patient, tooth, diagnosis, appointment and treatment.

3. Clinical Case History Templates

- General dentistry: chief complaint, HPI, dental history, medical history, examination, diagnosis and treatment plan.
- Pediatric dentistry: diet, oral hygiene, fluoride exposure, habits, caries risk, Frankl behaviour score, growth/development, trauma and pulpal diagnosis.
- Endodontics: pulpal/periapical diagnosis, percussion, palpation, cold/EPT, mobility, probing, radiographic findings, working length, file system, irrigant, sealer and obturation.
- Expandable specialty templates for prosthodontics, orthodontics, periodontics, oral surgery and implants.

4. Treatment Planning

- Select tooth + procedure and generate a structured treatment plan.
- Automatically calculate fee, estimated appointments, consent requirement, materials, laboratory requirement and expected completion.
- Patient-facing treatment presentation with simple explanations and treatment acceptance.

5. Digital Consent System

- Procedure-specific consent templates for RCT, extraction, surgery, implants, crowns, pediatric pulpectomy/SSC/pulpotomy, sedation, biopsy, orthodontics, aligners and prosthetic procedures.
- Patient reads, acknowledges and signs digitally.
- Capture patient, procedure, tooth, risks, alternatives, complications, declaration, dentist, signature and date/time.
- Consent must remain linked to the exact case/treatment.

6. Clinical Photos & Imaging

- Store intraoral/extraoral photographs, IOPA, OPG, CBCT, PDFs and scan files.
- Organize by Patient → Case → Tooth → Appointment.
- Before/after comparisons and easy retrieval.

7. Appointment Management

- Appointment types: consultation, RCT, crown preparation/delivery, extraction, implant, pediatric treatment, review and emergency.
- Automated confirmation, 24-hour reminder, same-day reminder, cancellation and rescheduling workflows.
- Calendar for dentists, consultants, chairs/rooms and clinic branches.

8. Consultant Doctor Management

- Maintain consultant specialties, schedules and availability.
- Automatically notify consultant of new cases and appointments.
- Share relevant patient/case information securely.
- Track consultant fee, cases completed, payments and outstanding amounts.

9. Consultant Network / Referral Ecosystem

- Clinics can maintain preferred pediatric dentists, endodontists, orthodontists, prosthodontists, periodontists and oral surgeons.
- Request → accept → schedule → treat → close case workflow.
- Potential future B2B specialist network across India.

10. Billing & Cash Sheet

- Treatment-level billing linked to tooth and case.
- Support advances, partial payments, discounts, refunds, outstanding balances and multiple payment modes.
- Daily cash reconciliation: opening cash + collections – refunds – expenses = closing balance.
- GST/tax fields configurable as required.

11. Case-Level Profitability — Key Differentiator

- Calculate actual revenue and direct cost for every case.
- Example: RCT revenue ₹5,000; materials, lab and doctor/associate costs are allocated to show contribution margin.
- Profitability reports by procedure, dentist, branch, specialty and time period.

12. Inventory Management

- Product master with batch, expiry, supplier, purchase price, GST, stock quantity and reorder level.
- Automatic stock deduction when materials are used in a case.
- Track wastage, damaged stock, expired stock and adjustments.
- Low-stock and expiry alerts.

13. Case-Wise Material Consumption

- Record material quantity and cost per procedure.
- Examples: rotary file allocation, gutta-percha, sealer, irrigants, anesthetic cartridges, needles and other consumables.
- Automatically feed material usage into case profitability.

14. Laboratory Management

- Track crowns, bridges, dentures, aligners, implant prostheses and surgical guides.
- Workflow: Sent → Received → Try-in → Modification → Delivered.
- Record lab, due date, fee, shade, material, tooth, instructions and digital files.
- Automatic overdue laboratory alerts.

15. Prescription Module

- Procedure/diagnosis-linked prescription templates.
- Drug, dose, frequency, duration and instructions.
- Allergy and medical-history alerts.
- Customizable dentist templates.

16. WhatsApp / Patient Communication

- Appointment confirmations and reminders.
- Payment reminders, post-operative instructions, review requests and recall messages.
- Buttons/links for booking, rescheduling and contacting the clinic.
- Maintain communication history.

17. AI Receptionist / Voice Bot

- Answer clinic calls, provide clinic information and check doctor availability.
- Book, reschedule and cancel appointments.
- Handle routine follow-up calls and reminders.
- Keep diagnosis, prescribing and clinical decision-making under dentist control.

18. Recall & Follow-Up Engine

- Automatically identify patients due for cleaning, pediatric recall, fluoride, sealants, crown review, implant review, orthodontic review and other recalls.
- Recall dashboard and automated outreach.
- Track contacted → responded → booked → completed.

19. No-Show & Follow-Up Intelligence

- Identify patients with repeated missed appointments.
- Use risk flags to support appropriate reminder or advance-payment policies.
- Track treatment plans that were diagnosed but not started.

20. Patient Portal

- Appointments, treatment plan, bills, prescriptions, X-rays/photos, consent, payments, instructions and recall dates.

- Simple patient-facing interface optimized for mobile.

21. Staff & Role Management

- Roles: owner, dentist, receptionist, assistant, accountant and consultant.
- Permissions by module and branch.
- Track attendance, salary, commissions, cases and productivity.

22. Clinic Owner Dashboard

- Revenue, expenses, collections, outstanding payments and contribution.
- New vs returning patients, treatment acceptance, no-shows and appointment utilization.
- Revenue by treatment, dentist, branch and referral source.
- Marketing ROI and patient acquisition analytics.

23. Treatment Acceptance Analytics

- Measure diagnosed cases vs accepted cases.
- Acceptance by treatment: RCT, crown, implant, orthodontics, pediatric procedures, etc.
- Identify high-value treatments where patients are dropping off.

24. Multi-Clinic Management

- One owner account can manage multiple clinics/branches.
- Separate branch dashboards plus consolidated group dashboard.
- Branch-specific staff, inventory, fees, expenses and reports.
- Database architecture should support multi-tenant/multi-branch use from Day 1.

25. Security & Audit Trail

- Role-based access and strong authentication.
- Audit who accessed, created, modified or deleted important records.
- Timestamp consent, prescriptions, billing changes and clinical notes.
- Design privacy/security for Indian data-protection requirements and applicable healthcare record obligations.

26. AI Clinical Documentation

- Dentist speaks clinical notes; AI converts them into structured case documentation.
- Suggested chief complaint, history, findings and treatment notes.
- Dentist reviews and approves before saving.
- Clearly distinguish AI assistance from dentist-authored clinical decisions.

27. AI Clinical Decision Support — Future

- Potential decision-support for structured symptoms/findings and radiographic observations.
- AI radiograph assistance may flag possible findings such as caries, periapical changes, bone loss, impacted teeth or restorations.
- Always require dentist verification; do not position the system as autonomous diagnosis.

28. Expense Management

- Track rent, salaries, electricity, internet, materials, lab, marketing, equipment, maintenance and consumables.
- Connect expenses to branch and, where appropriate, individual cases.
- Generate monthly profitability and cash-flow reports.

29. Equipment & AMC Management

- Equipment register: chair, compressor, autoclave, X-ray/RVG, microscope, endomotor, scanner, implant motor and more.
- Purchase date, warranty, AMC, service history and next-service date.
- Automatic maintenance reminders.

30. Sterilization & Infection Control

- Autoclave cycle log with date/time, operator and cycle number.
- Sterilization indicators and biological indicator records.
- Instrument-pack tracking and maintenance.
- Daily infection-control checklist and biomedical-waste checklist.

31. Referral & Marketing CRM

- Track referral source: existing patient, doctor, clinic, Google, Instagram, website, Practo, walk-in, camps and other channels.
- Calculate patients and revenue generated by source.
- Create follow-up lists for untreated and overdue patients.

32. AI Clinic Copilot

- Natural-language questions over clinic data.
- Examples: 'How much did I earn from pediatric cases last month?', 'Show pending crowns', 'How much did I spend on endodontic materials?', 'Which treatment has the highest margin?'
- Use strict role-based access so AI only exposes data the user is permitted to see.

Recommended Product Architecture

The core data model should revolve around one connected chain:

PATIENT → CASE → TOOTH → DIAGNOSIS → TREATMENT PLAN → APPOINTMENT → CONSENT
→ MATERIALS → LAB → TREATMENT → BILL → PAYMENT → FOLLOW-UP → RECALL →
PROFITABILITY

MVP — Build This First

Priority	Module
1	Patient management + medical history
2	Dental chart + case sheet
3	Treatment planning
4	Appointments + reminders
5	Billing + cash sheet
6	Digital consent
7	Inventory
8	Case-wise material costing
9	WhatsApp communication
10	Basic clinic dashboard

V2 — Clinic Operations

Laboratory management • consultant management • recall engine • expenses • advanced profitability • staff productivity • multi-clinic management • patient portal.

V3 — AI Layer

AI receptionist/voice bot • voice-to-case documentation • AI clinic analytics • treatment follow-up intelligence • carefully controlled clinical decision support.

Five Core Differentiators

- 1. Case-level profitability** — Know exactly what every dental procedure costs and earns.
- 2. Intelligent inventory** — Materials are automatically deducted from actual cases.
- 3. AI receptionist** — Calls, bookings, reminders and routine patient communication.
- 4. Consultant ecosystem** — Connect clinics and visiting specialists with shared scheduling and case workflows.
- 5. AI clinical documentation** — Dentist speaks; the system converts it into structured clinical notes for review.

Critical Development Principle

Do not build 40 modules at once. Validate the MVP with real dental clinics first. However, design the underlying database from Day 1 for multi-clinic, multi-doctor, tooth-level, case-level, inventory and financial relationships. Retrofitting these relationships later can be expensive and technically difficult.

Product Positioning

Do not market it as just another dental appointment/billing application. Position it as a **Dental Clinic Operating System** that connects clinical care with operations and profitability.